



CASE STUDY SERIES

From Zero to a Governed System.

*How a non-trader, non-engineer used Claude to research, build, reject, rebuild and govern an automated trading system — and what it reveals about governing any autonomous **AI decision system.***

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FOREWORD

Why this case study exists.

I am not a trader. I am not a quant. I am not a developer. This is precisely why this document matters.

This playbook documents something I did not set out to build. It started as a single, slightly suspicious prompt — a request to copy whatever indicators “really good” public trading tools were supposedly using and install them on my Bitcoin chart. It ended, several weeks later, as a fully governed, auditable, paper-trading decision system with a frozen and validated model, a live risk-management engine, a committee-style review process, and an executive governance dashboard.

Nothing about that journey required me to already know quantitative finance, Pine Script, n8n, or governance architecture. What it required was a working method: question the premise, test rather than assume, demand evidence before believing a result, and refuse to let a system go live until it could explain itself.

That method is the actual subject of this case study. The trading system is the worked example — it happens to be vivid, numerate, and falsifiable, which makes it a better teaching tool than most business processes. But the underlying pattern is the one I now use with every client building or governing an AI system: define the mandate, test the model honestly, reject what doesn’t hold up, freeze what does, and wrap the whole thing in a layer that can answer “why was this decision allowed?” at any point in the future.

If you read only one sentence from this document, read this one: the most valuable thing I built was not a trading strategy. It was a record of how I learned not to trust it blindly — and a system that makes that scepticism permanent, automated, and auditable.

HOW TO READ THIS

If you want the trading story: Chapters 2 and 3 walk through every version of the strategy, in plain English, including the versions that failed.

If you want the governance story: Chapters 4 and 5 are the heart of the case study — how monitoring became governance.

If you want the AI Fluency takeaway: Chapter 6 makes the connection explicit, and is the chapter I would hand to a client.

CHAPTER 1

The wrong question, and the right one.

Every AI-assisted project starts with a prompt. The prompt you write is usually a confession of what you actually believe — whether or not you realise it.

My first instinct was a completely ordinary one: find the people who already do this well, copy what they do, and apply it. The prompt I gave Claude said as much — find the “very public tools” that day traders use, identify their indicators and moving averages, install them on my Bitcoin chart, and let me watch it play out.

It is worth sitting with why that prompt feels reasonable. It mirrors how most of us learn most things: find an expert, copy their method, adapt it. The problem is that this particular shortcut runs directly into one of the best-documented facts in retail finance — the overwhelming majority of day traders lose money, transaction costs erode most of what little edge exists, and a credible, durable, publicly-marketed edge in fast trading essentially does not exist for retail participants. Asking an AI to find that edge and copy it was asking it to find something the data says is not reliably there.

The structural flaw was bigger than the statistics.

The deeper problem with the original prompt had nothing to do with markets specifically. It asked for an unnamed authority to be trusted uncritically, for its method to be installed without being tested, and for the result to be evaluated by watching it — visually, subjectively, for a short period. No success criteria. No falsification step. No risk constraints.

That pattern is not unique to trading. It is the same pattern that produces bad AI pilots in organisations: find a vendor with an impressive demo, install their black box, watch it for a few weeks, and decide it “seems to be working.” The fix in both cases is identical — replace “install it and watch” with “build it, test it against real history, and read the numbers.”

The better approach: one simple, transparent, testable strategy.

Rather than installing somebody else’s opaque indicator stack, the redesigned approach rested on four disciplines that recur throughout this entire case study, not just at the start:

- A simple, transparent core. A trend-following regime filter — the kind of approach with the most academic and institutional support on an asset like Bitcoin — rather than a stack of unexplained indicators.
- One confirmation signal, one risk tool. Momentum (RSI) to confirm the trend has strength, and volatility (ATR) to size stops — deliberately not more, because additional indicators mostly add ways to curve-fit a backtest, not genuine edge.
- Backtest before believing. Every claim had to be proven against real historical data inside TradingView’s Strategy Tester — numbers, not vibes.



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- Fixed, explicit risk rules. A defined position size and a defined stop on every single trade, because in a trend-following system, controlling the losses is where the entire edge is actually earned.

THE REFRAME IN ONE LINE

The original prompt asked Claude to find an authority and copy it. The working method that replaced it asked Claude to build something, prove it against ten years of real data, and only then decide whether to trust it. Same screen. Completely different epistemics.

CHAPTER 2

Building the trading logic, one honest test at a time.

For a reader with no trading background: this chapter explains, in plain language, what the strategy actually does and why each design choice was made — including the early version that did almost nothing, which turned out to be the most important result of all.

The starting structure: a regime filter.

The first version of the strategy rested on one idea that shows up across almost all serious trend-following systems: only take trades when the medium-term trend is above the long-term trend. In practice this meant comparing two moving averages of price — a 50-day average and a 200-day average. When the 50-day average sits above the 200-day average, the market is treated as being in a “bull regime” and the strategy is allowed to look for trades. When it sits below, the strategy stands aside entirely.

On top of that regime filter sat two more pieces. RSI (Relative Strength Index) measured whether momentum was still healthy at the moment of a potential entry. ATR (Average True Range) measured how much the asset was currently moving, and was used to size the protective stop-loss — a wider stop in a volatile period, a tighter stop in a calm one.

The first real test: three months, \$1,000, and a result of nothing.

The first thing the strategy was asked to do was simulate \$1,000 invested over the prior three months, applying its rules mechanically. The result: \$1,000 stayed \$1,000. Zero trades. Meanwhile, simply buying and holding Bitcoin over the same window would have lost roughly 9.7%, as the price fell from the mid-\$70,000s to the mid-\$60,000s.

This looked, at first glance, like a failure. It was actually the single most important validation in the entire project. The regime filter had correctly identified that the 50-day average was still miles below the 200-day average — the scar tissue of an earlier crash from above \$120,000 — and refused to buy into a market still being signalled as untrustworthy. A strategy that produced zero trades and zero losses while the market fell almost 10% was not broken. It was doing exactly what a defensive, regime-aware system is supposed to do: nothing, when nothing is the correct answer.

THE LESSON WORTH KEEPING

A backtest that shows excitement during a falling market is the warning sign. A backtest that shows boredom is usually the honest one. Judging any model — trading or otherwise — on a

short, quiet window tells you almost nothing; it takes enough trades and enough different market conditions before the numbers mean anything at all.

Diagnosing “why didn’t it trade” before changing anything.

Rather than immediately loosening the rules, the next step was a forensic diagnosis: which exact condition was blocking trades, and was that condition working as intended or simply broken? The answer, confirmed against the actual price data rather than guessed at, was that the regime filter was the sole cause — and it was working correctly. A secondary, genuine bug was also found and logged for a later fix: the protective stop calculation referenced a value that did not yet exist on the very first bar of a new trade, silently leaving early trades unprotected. Diagnosing precisely, before redesigning, is what separates an upgrade from a guess.

Redesigning for more opportunities, on a faster clock.

Three deliberate changes produced the next working version of the strategy. First, the timeframe moved from daily candles to four-hour candles — the same logic, but checked six times more often, so the long-term average could respond to a genuine trend change in roughly a month rather than the better part of a year. Second, the entry trigger changed from “wait for price to cross above the average” (which happens rarely) to “enter when price pulls back close to the average during an established uptrend” (which happens far more often, while keeping the same underlying logic: price respecting its trend). Third, the exit logic was anchored to the actual low of the pullback candle — the real point at which the trade idea is proven wrong — rather than to an arbitrary fixed distance.

Every one of these changes was paired with an explicit non-repainting safeguard: signals were only ever confirmed on a fully closed, unchangeable candle, and orders only ever executed at the next candle’s open. This single discipline is what stops a backtest from quietly lying to you by using information that would not have been available in real time.

The first full, honest verdict.

Once the redesigned strategy had enough trade history to mean something — 5.5 years of real four-hour Bitcoin data — the picture was genuinely encouraging on four counts and genuinely concerning on one.

Metric	Result	Read
Profit Factor	2.31	Strong — gross wins over 2× gross losses
Win Rate	47.8%	Comfortably above the 33% needed at 2:1 reward-to-risk

Metric	Result	Read
Net result vs buy-and-hold	+\$7,531 over 5.5 years	Genuine outperformance, not just riding the asset's rise
Maximum drawdown	35.9%	Failed the 20% target — the headline problem
Number of trades	23	Workable, but thin for full statistical confidence

A 35.9% drawdown is not a rounding error — it is the kind of loss that causes real people to abandon a system at exactly the wrong moment. The fix was not a redesign of the entry logic. It was a risk-sizing change: reducing the amount risked on each individual trade from 2% of capital to 1%. Profit factor is a ratio of wins to losses and is untouched by position size; only the magnitude of the swings changes. That one change, on its own, cut maximum drawdown nearly in half.

A second, harder audit — because “it backtests well” is not enough.

Before trusting the strategy further, it was put through an eight-question audit focused entirely on reliability rather than performance: was the non-repainting logic actually sound; was the position-sizing safe for an asset as volatile as Bitcoin; did the stop-loss and cooldown logic use the right underlying data (a candle's low, not its closing price) to decide whether a stop had truly been hit? The audit found three genuine bugs — all fixed — and confirmed several design choices were already correct. This is the version that became the strategy's stable foundation: max drawdown reduced from 35.9% to 8.55%, profit factor improved to 2.56, and trade count nearly doubled to 44, giving the statistics real weight.

	Before audit	After audit
Max drawdown	35.92%	8.55%
Profit factor	2.31	2.56
Trade count	23	44
Largest single loss	\$1,530	\$476

Two things about this version are worth naming explicitly, because they say more about the method than the metrics do. First, every change made was a correction to an identified, evidenced failure mode — not a tweak made because it happened to improve the backtest. That distinction is the entire difference between robustness and curve-fitting. Second, the strategy was then run, completely unmodified, on Ethereum and on Gold. It stayed profitable on both — strong evidence that the logic was capturing something real about how trending markets behave, rather than something that happened to fit Bitcoin's specific history.

CHAPTER 3

Adding intelligence — and learning when to reject it.

This is the chapter I would show a client who thinks AI governance is about approving the clever idea. It is just as much about having the discipline to reject it.

Where the idea came from.

A YouTube video on quantitative trading introduced the idea of a Markov regime model — a way of describing the market not as simply “bull” or “bear” but as a set of probabilities: given today’s short-term trend, what is the probability tomorrow continues in the same direction? The video overstated its case in places — framing quant trading as a secret retail traders are denied, and treating a serious technique (a Hidden Markov Model) as something that installs itself in minutes. But the underlying mathematics was legitimate, and one specific idea was genuinely worth testing: replacing a binary regime filter with a probability-weighted one.

The discipline that mattered here was refusing to simply bolt the new idea on. Before any code was written, three questions were answered in writing: does this genuinely complement the existing strategy, does the existing strategy already do something similar, and — critically — what would prove this is a real improvement rather than a coincidence?

Version 5: the Markov gate, and why it was rejected.

The first implementation added the Markov probability as a hard gate — a trade could only fire if both the existing trend logic and the new probability signal agreed. An investment-committee-style review was defined in advance, with eight measurable pass/fail criteria, before a single live result was read. That sequencing matters: criteria set after seeing the result are not criteria, they are excuses.

	v4 (existing)	v5 (Markov gate)	Verdict
Profit Factor	2.558	2.658	Marginal improvement — short of the +0.20 bar set in advance
Max Drawdown	8.55%	9.66%	FAILED — the primary objective got worse, not better
Win Rate	34.1%	45.2%	Improved sharply
Trade Count	44	31	Below the statistical floor the committee required
Sharpe Ratio	0.187	0.158	FAILED — risk-adjusted return declined

Version 5 was rejected. Not adjusted, not quietly kept “just in case” — rejected, with the existing version reinstated as the approved baseline. The committee’s own rule made this automatic: if a change fails its primary stated objective, it is rejected regardless of how good it looks elsewhere. Drawdown was the entire reason the Markov layer was being explored, and it got worse.

The paradox that explained why — and pointed to the real fix.

The data held something genuinely counter-intuitive: the win rate improved by eleven percentage points at the same time as the drawdown got worse. Both were true. The explanation came from looking at which trades the filter had actually removed. It had screened out many small, fast losses — the everyday cost of trading through choppy, uncertain periods — but it had let through fewer, slower-developing losing trades that took longer to play out and cost more on average. The filter was making the system look more accurate while leaving the most damaging individual losses untouched.

This is exactly the kind of finding that a single headline metric — “win rate improved!” — would have hidden. It only surfaced because the evaluation method demanded multiple criteria, set in advance, with an explicit instruction not to optimise for the metric that looked good.

Version 6: the same signal, used correctly.

The insight from the rejection was the design for the next version: if the Markov signal contains real information about trade quality — which the win-rate jump proved it does — the correct use of that information is not a yes/no gate. It is proportional position sizing. Trade normally when the signal is weak; risk more capital only when the signal is strongly in agreement with the existing trend logic.

- Signal above 0.70 (strong agreement) → risk 1.00% of capital, the maximum.
- Signal 0.50–0.70 → risk 0.75%.
- Signal 0.35–0.50 → risk 0.50%.
- Signal 0.20–0.35 → risk 0.25%.
- Signal below 0.20 → risk a minimal 0.10%, rather than blocking the trade entirely.

Crucially, every single entry and exit rule from the existing, validated strategy stayed completely unchanged. Only the amount of capital placed behind each trade moved, in proportion to how strongly the new signal agreed with the old one.

The result resolved the paradox directly. Trade count held at exactly 44 — nothing was filtered out, so nothing was lost from the sample. Win rate and profit factor held essentially flat. And maximum drawdown fell from 8.55% to 5.11%, a 40% reduction — because the trades that turned out to be the most damaging were now sized down, rather than fully exposed, when the signal had been weak going in.



WHY THIS IS THE MOST IMPORTANT CHAPTER FOR AI GOVERNANCE

An AI signal that improves one metric is not automatically a good idea. The committee process — criteria set in advance, multiple dimensions scored simultaneously, an explicit instruction not to optimise for the headline number, and a willingness to reject a result that initially looks better — is the exact discipline that separates genuine model improvement from a more convincing-looking mistake. The same process applies whether the model is choosing trades or approving invoices.

CHAPTER 4

Freezing the model and building the governance layer.

Once a model has earned trust through evidence, the next discipline is refusing to keep fiddling with it — and building the structure that proves, continuously, that it is still behaving the way it was approved to behave.

Calibrating the risk schedule — and finding the efficient frontier.

Position sizing by signal strength (Chapter 3) raised an obvious follow-up question: what exact percentages should sit at each tier? Four candidate schedules were tested, all keeping the entry and exit logic completely untouched and varying only the risk percentages assigned to each signal tier.

Schedule	Tiers (high → low signal)	Profit Factor	Max Drawdown
A	1.00 / 0.75 / 0.50 / 0.25 / 0.10	2.574	5.11% / 6.14% of capital
B	1.00 / 0.80 / 0.60 / 0.40 / 0.20	2.761	5.93%
C	1.00 / 0.90 / 0.70 / 0.50 / 0.25	2.817	6.38%
D	1.00 / 0.85 / 0.70 / 0.55 / 0.40	2.566	7.59%

Schedule C scored highest on a composite weighting of drawdown, annual return, risk-adjusted return and profit factor — and produced the best profit factor of any version built across the entire project, while keeping drawdown 25% below the original audited baseline. This is the version that became the approved, frozen strategy: v6C Schedule C.

Validation, not development: the three tests that earned the freeze.

Once Schedule C was selected, the explicit instruction to Claude changed. No more new ideas, no more parameter exploration — validation only, with three specific tests run against the unmodified, frozen configuration.

- Full-history test. Run across the maximum available real four-hour Bitcoin data, spanning a complete bull market, a severe 77% bear-market crash, and a recovery. The strategy took zero trades during the entire 2022 collapse — the regime filter correctly sat out the worst crypto year on record rather than trying to catch a falling market.
- Out-of-sample reasoning. Performance on the period after the strategy was originally calibrated needed to hold up on data the design choices hadn't been shaped around — the

discipline equivalent to a train/test split, and the strongest available defence against a backtest that merely memorised its own history.

- Multi-asset robustness. The identical, unmodified strategy was run on Bitcoin, Ethereum and Gold. It was profitable on all three — strong evidence that the logic is a genuine trend-following framework rather than a Bitcoin-specific coincidence. Gold in particular produced an exceptionally clean result (profit factor above 3, drawdown under 3%), confirming the risk architecture itself was sound even on a structurally calmer asset.

The committee’s formal decision was Limited Capital Approved — not yet “production ready,” specifically because one test (a fully interactive out-of-sample run inside the platform’s own backtester) could not be completed through automation in that session and required a manual confirmation step. That is itself a governance principle worth naming: the system declined to award itself a higher confidence rating than the available evidence supported, even though the analytical case was strong.

From monitoring to governance: the missing question.

At this point the project had a validated model and a clear set of guardrails. What it didn’t yet have was a way to answer one specific question, at any point in the future, for any single decision: why was the system allowed to act?

Standard logs — a record of signals, a record of trades, a record of governance events — answer what happened, when it happened, and what the outcome was. They do not answer why the system was authorised to act in the first place. That distinction is the line between monitoring and governance.

The Decision Basis Record.

The fix was a structural one: a record, attached to every single trade, of the specific conditions that had to be true for that trade to be permitted — captured at the moment of entry, not reconstructed afterwards from memory.

Trade ID	Signal	Risk Tier	Status	Reason
BTC-001	0.82	1.00%	Approved	Strong bull regime, profit factor within mandate
BTC-002	0.41	0.50%	Approved	Medium-conviction signal, reduced exposure
BTC-003	0.23	0.25%	Approved	Weak signal, minimum exposure applied

Each record traces a trade back to an authority reference — the specific committee approval and strategy version that sanctioned it — so that six months later, the answer to “why was Trade 17 taken?” is not a reconstruction. It is a read of data that was captured before the trade

was ever placed: the regime was confirmed bullish, model drift was within tolerance, drawdown was within limit, the consecutive-loss count was below its threshold, the Markov signal was 0.67, and the resulting risk tier was 0.75%.

One line, sitting quietly at the bottom of the dashboard, changes the meaning of everything above it: Strategy frozen at v6C Schedule C. Without that statement, what has been built is a dashboard. With it, what has been built is a controlled system — the equivalent of an approved model version, an approved policy version, an approved risk methodology. It is the line that creates accountability.

THE GOVERNANCE STACK, END TO END

Mandate Status — is the system authorised to operate, right now, at a glance?

Mandate Compliance — are the individual controls passing?

Governance Controls — are stated limits being respected?

Model Drift Monitor — is live behaviour diverging from the approved, validated baseline?

Validation Evidence — is reality still matching the approved model?

Audit Logs — what happened, and when?

Decision Basis Record — why did it happen, and who authorised it?

Committee Analyst — a read-only interpreter that may explain the evidence, and may never change the system.

CHAPTER 5

The operating system: how the pieces connect.

A validated strategy and a governance philosophy are only as real as the system that enforces them automatically, every single day, without anyone needing to remember to check.

The build brief: operationalise, don't improve.

The instruction given to Claude at this stage was deliberately narrow, and worth quoting because the discipline is the point: build the operational system, not a better strategy. The risk in handing a frozen, approved model to an AI assistant is that it will “helpfully” start improving it — quietly proposing a v7, adjusting a risk rule, redesigning a workflow step. The brief explicitly forbade all of that. The frozen strategy — v6C Schedule C, with its five-tier risk schedule — was treated as a fixed input, not a variable.

The five-component architecture.

Component	Role
TradingView	Single source of truth for price data and signals. Runs the frozen strategy and fires webhook alerts. Touches no other logic.
n8n	The orchestration and enforcement engine. Validates every incoming signal, calculates position size from the Markov risk schedule, writes every record to the journal, and is the only component permitted to enforce a governance block.
Google Sheets	The persistent, auditable journal of record — separate tabs for trades, open positions, daily metrics, the Markov signal log, every governance event, and the frozen configuration itself.
Governance dashboard	The executive view — mandate status, live controls, drift monitor, audit trail and Decision Basis Record, built to be read correctly from a phone in under five seconds.
Telegram	The human-in-the-loop channel. Every signal, every exit, every governance breach and every scheduled report arrives here — nothing happens silently.

The governance rules that actually run, automatically.

Control	Threshold	Behaviour
Drawdown warning	8% of capital	Soft alert — trading continues, a human is notified to watch
Drawdown halt	12% of capital	Hard stop — no new entries until a human reviews and explicitly resumes
Maximum risk per trade	1% of capital	Hard ceiling, enforced regardless of signal strength

Control	Threshold	Behaviour
Consecutive loss review	5 in a row	Triggers mandatory human review — the backtested worst case was exactly five
Strategy integrity	Must equal v6C Schedule C	Any parameter drift from the frozen configuration blocks the trade and flags governance

The parameter-freeze check deserves particular attention because it is the mechanical enforcement of the entire “do not optimise” discipline. Every time a signal arrives, the orchestration layer reads the approved configuration and confirms the incoming risk tier matches one of the five sanctioned values exactly. Any deviation — even an accidental one, even a well-intentioned one — is blocked and logged as a governance event before it ever becomes a trade. The system cannot be quietly tweaked. It can only be formally amended.

Designing the dashboard for a human, not a trader.

The dashboard went through several genuine redesigns, and each one is a small lesson in interface design for any AI oversight system. The first version was, in its own honest description, a dark-themed trader’s dashboard — hard to read, P&L-centric, built around the question “how much money is this making?”

The questions that actually matter to someone responsible for oversight turned out to be different: is the system authorised to act right now, are its controls within mandate, and is its behaviour still matching what was validated? The redesign reordered the entire page around those questions. A single mandate-status indicator — a coloured light reading Within Mandate, Review Required, or Mandate Breached — sits at the very top, deliberately legible enough to be read correctly from a phone in a few seconds. A five-control governance scorecard sits directly below it, reducing the entire system’s health to one number an executive can hold in their head: “5 of 5 controls passing.” The equity curve — the most trader-centric element of all — was demoted from hero position to a supporting panel relabelled “validation evidence.”

The vocabulary changed deliberately throughout: a profit-and-loss log became an execution audit log; a signal feed became a signal audit trail; a chat box for asking questions became a Committee Analyst, explicitly read-only, explicitly forbidden from recommending any change to the frozen strategy.

CHAPTER 6

What this is really a case study in.

Replace the trading vocabulary with operational vocabulary, and what remains is a working reference architecture for governing any autonomous decision system.

The substitution that reveals the real subject.

This project's language	The general governance equivalent
Profit Factor	Model accuracy / decision quality
Drawdown	AI risk exposure
Trade Log	AI decision log
Markov signal strength	Model confidence score
Strategy frozen at v6C Schedule C	Approved model version / approved policy version
Mandate Status	Is the system authorised to act, right now?
Decision Basis Record	Reconstructable justification for any single automated decision
Committee Analyst (read-only)	An AI interpreter permitted to explain, never to act or override

Imagine, instead of a trading system, an AI agent approving invoices. An auditor asks: why did the AI approve this payment? “The AI thought it was fine” is not an answer a regulator, a board, or an ISO/IEC 42001 auditor will accept. “The approval occurred because the confidence score exceeded the defined threshold, the vendor was on the approved list, the risk score was below the limit, human oversight was not required under the policy in force, and policy version 2.3 was active at the time” is exactly the same structure as the Decision Basis Record built for a Bitcoin trade. The pattern transfers completely.

The seven disciplines, in the order they actually have to happen.

1. Question the premise before building anything. The very first prompt in this project was wrong, and recognising why it was wrong — rather than simply executing it — was the single highest-leverage decision in the entire case study.
2. Build something simple and transparent before anything sophisticated. A two-moving-average regime filter, RSI and ATR. Complexity was earned through evidence, never assumed up front.
3. Test honestly, including the test that produces a boring or disappointing result. The three-month, \$1,000 simulation that returned exactly \$1,000 was the most valuable single result in the project, precisely because it was unflattering and correct.



4. Audit for reliability, separately from auditing for performance. Three real bugs were found by asking “is this trustworthy” rather than “is this profitable.” A system can backtest brilliantly and still be unsafe.
5. Set success criteria before seeing the result — and be willing to reject. The Markov-gated version (v5) looked better on its headline metric and was rejected anyway, because the criteria that mattered most had been defined in advance and it failed them.
6. Freeze what has earned trust, and make the freeze mechanically enforced. v6C Schedule C is not a suggestion sitting in a document — it is a parameter the orchestration layer actively checks on every single signal, and blocks on any deviation.
7. Build the layer that can explain itself, automatically, to someone who wasn't there. Logs answer what happened. The Decision Basis Record, the Drift Monitor and the Mandate Status answer why it was allowed to happen — which is the question every real governance regime ultimately has to answer.

IF YOU TAKE NOTHING ELSE FROM THIS CASE STUDY

You do not need to already understand the subject matter, and you do not need to already know how to connect the tools, to end up with a working, governed system. What you need is the discipline to keep asking one question at every stage: how would I know if this were wrong? Build the thing that answers that question, and govern accordingly.

CLOSING

Three commitments for anyone starting from zero.

This case study began with a prompt I am slightly embarrassed by, and ended with a governance architecture I would defend in front of an audit committee. The gap between those two points was not expertise. It was method.

1. Replace “copy the expert” with “build, test, and read the evidence.”

Wanting to learn from people who are good at something is not the flaw. Wanting to install their conclusion without testing it for yourself is. The discipline of building a simple, transparent version of an idea and proving it against real evidence — rather than trusting a claimed result — is available to anyone, regardless of starting expertise, and it is the single most transferable habit in this entire document.

2. Define what would count as failure before you look at the result.

The Markov-gated strategy was rejected, decisively, because the criteria for success had been written down before the live numbers were read. That sequencing is the entire difference between genuine evaluation and finding a way to be pleased with whatever you got. It applies as much to an AI vendor pitch as it does to a trading model.

3. Build the system that can explain itself — don't just build the system that performs.

A working model is not the same thing as a governed one. The work that turned this project from a trading experiment into a genuine governance case study was almost entirely the work that happened after the model was already good: the Mandate Status, the Drift Monitor, the Decision Basis Record, the frozen parameter check, the read-only analyst. None of that work makes the system perform better. All of it makes the system defensible — which, for any AI system operating with real consequences, is the actual point.

*This is the version of AI fluency I work with clients to build: **not technical mastery of every tool, but the judgement to question, test, reject, freeze, and govern — applied consistently, regardless of the domain.***

ABOUT

The AI Fluency Company.

Zoë François is the founder of The AI Fluency Company, a Cape Town-based consultancy focused on AI governance, strategic AI readiness, and responsible AI adoption for the South African mid-market. She trained as a chemical engineer and spent two decades in commercial leadership in industrial sectors, including roles at **Sappi** and **Consol Glass**, and holds the Oxford Saïd Business School AI Programme certificate.

Her services span AI Fluency Training, Strategy Workshops, AI Opportunity & Risk Diagnostics, and ISO/IEC 42001 Readiness engagements, with a particular focus on compliance-minded mid-market businesses.

This case study is offered as a worked example of AI-augmented build discipline — not as trading advice, financial advice, or a recommendation to trade. The system described is a paper-trading governance prototype and has not been deployed with live capital.

TAKE THE NEXT STEP

Take the AI Fluency Assessment for a maturity benchmark: theaifluencyco.com

Book a 30-minute conversation to discuss what a governance control tower could look like for your own AI initiatives — no charge for the first call.

Ask about the companion Executive Series — the Executive's Guide to AI Governance and the Executive's Guide to the AI Product Lifecycle — for the disciplines that sit behind this case study.

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